Table 3: **Object-based variants, performance comparison.** The values correspond to the number of tasks solved by each method, out of 50; higher is better and the top-performer is bolded.

Object-based encoding	Few-shot		In-context Few-shot w/ CoT	
	GPT-3.5	GPT-4	GPT-3.5	GPT-4
Object Descriptors	9	16	5	20
Object Descriptors with Edge	5	18	4	12
Object JSON	8	21	6	**23**
Object JSON with Edge	5	19	4	16

Table 4: **Results for object-based approach.** In addition to the caption of Table 2, the numbers in parentheses are the increase in the percentage of tasks solved with the object-based approach compared to the direct-grid approach of Table 2. The number of tasks in each column is 50, so a 2% increase means that a given method has solved one more task.

LLM	ARC Subset	Move 1 Pixel	Move 3 Pixels	Move Dynamic	Recolor by Size	Denoise	Fill		Move		Pile	
							H	V	H	V	H	V
GPT-3.5	6 (+8%)	39 (+58%)	14 (+14%)	7 (+2%)	21 (+38%)	36 (+46%)	17 (+30%)	15 (+30%)	1 (+2%)	0 (-)	7 (+10%)	10 (+20%)
GPT-4	23 (+20%)	50 (+34%)	49 (+74%)	37 (+52%)	40 (+52%)	50 (+20%)	48 (+4%)	49 (+96%)	21 (+18%)	20 (+40%)	42 (+20%)	37 (+74%)
LLAMA2-13B	6 (+12%)	29 (+42%)	9 (+12%)	7 (+6%)	4 (+8%)	17 (+26%)	2 (+4%)	3 (+6%)	0 (-)	0 (-)	1 (+2%)	0 (-)
LLAMA2-70B	7 (+10%)	39 (+56%)	17 (+26%)	6 (+2%)	8 (+16%)	19 (+36%)	6 (+12%)	5 (+10%)	0 (-)	0 (-)	2 (+4%)	0 (-)

4.2 Results

We leveraged the prompting methods outlined in Section 2.2 in combination with our proposed object-based textual representations, replacing the direct-grid encoding. The results, presented in Table 3, show a marked improvement, with the success rate increasing from 13/50 tasks to 23/50 tasks on the ARC subset. Table 4 also shows that the previously observed performance gap between horizontal and vertical tasks in Table 2 is eliminated with the object abstraction, confirming our hypothesis that GPT's challenges with object cohesion in non-sequential text were the root cause. The orientation of objects becomes inconsequential, as desired. An even bigger performance boost is observed for the 1D-ARC, where GPT-4 achieves 50/50 on some task types. LLAMA-2 Touvron et al. (2023) models also exhibit an increase in performance for the object-based method.

On the complete ARC training set, the new object-based method allowed GPT-4 to solve 97 out of the 400 tasks, up from 81 previously. Likewise, on the Mini-ARC dataset, performance improved to 45 out of 149 tasks, up from 35. These results underscore the value of augmenting the LLM with an external tool that provides an appropriate representation, particularly when it comes to ARC tasks.

4.3 Analysis

We conducted the same solvability regression analysis from Section 2.4, observing the same correlations between task complexity attributes and solvability. Intriguingly, the models' performance was observed to decline when edge information was integrated into the representation. This unexpected result suggests that the influx of excessive information might overwhelm GPT, resulting in diminished performance. This discovery underscores the need for future research to find an optimal balance between supplying adequate contextual information and avoiding information overload.

Furthering our analysis, we performed a similar examination of the tasks correctly solved by GPT under the object-based approach. Out of the 23 tasks that produced the correct output, an impressive 20 tasks

exhibited correct reasoning (See Appendix D). This significantly improved reasoning performance underscores the impact of effective object abstraction on GPT's reasoning abilities. Figure 5 additionally showcases GPT's reasoning when prompted using the object-based approach for a task where the direct-grid approach initially fell short in providing accurate reasoning.

5 Related work

Prompting methods for LLMs This is a very active area of development (Qiao et al., 2022; Huang and Chang, 2022). CoT prompting is introduced in (Wei et al., 2022), providing LLMs with intermediate reasoning steps leading to improved performance on some complex reasoning tasks. Extending this, (Kojima et al., 2022) demonstrated LLMs' potential as zero-shot reasoners by incorporating a "Let's think step by step" phrase in the prompt. This approach notably enhanced accuracy on various reasoning tasks, thus hinting at untapped zero-shot capabilities within LLMs that can be leveraged through simple prompting techniques.

Augmented LLMs The survey by Mialon et al. (2023) emphasizes the potential of augmentation in overcoming limitations of a pure LLM approach. "Toolformer" self-learns to use external tools via APIs, significantly improving zero-shot performance across various tasks (Schick et al., 2023). Program-Aided Language models (PAL) (Gao et al., 2022) combines the strengths of LLMs with a Python interpreter to accurately solve some reasoning tasks.

Solvers for the ARC Since the introduction of the ARC in 2019, various methods have been proposed to address it. A powerful DSL coupled with an efficient program synthesis algorithm has the potential to solve the ARC, as initially proposed in (Chollet, 2019). Notable examples include the Kaggle challenge's (Kaggle, 2020) winning solution, which utilized a manually-created DSL and DAG-based search for program synthesis (top quarks, 2020). Other high-ranking Kaggle participants followed similar strategies (de Miquel Bleier, 2020; Golubev, 2020; Liukis, 2020; Penrose, 2020). Fischer et al. (2020) employed a Grammatical Evolution algorithm within their chosen DSL, while (Alford et al., 2021) utilized the DreamCoder program synthesis system (Ellis et al., 2020) to derive abstractions from a basic DSL and compose solutions for new tasks through neural-guided synthesis. More recently, ARGA (Xu et al., 2023) was proposed as an object-centric framework that represents images using graphs and tree search for a correct program in a DSL based on the abstracted graph space. Alternative approaches for the ARC challenge have also been explored. The Neural Abstract Reasoner, a deep learning method, achieved success on a subset of ARC tasks (Kolev et al., 2020). Assouel et al. (2022) devised a compositional imagination technique to generate unseen tasks for enhanced generalization. (Ferré, 2021) focused on an approach based on descriptive grids. However, these alternatives have not yet surpassed state-of-the-art results.

ARC-like datasets have been introduced to tackle the ARC's complexity. The Mini-ARC (Kim et al., 2022), a 5×5 compact version of the ARC, was generated manually to maintain the original's level of difficulty. The Sort-of-ARC (Assouel et al., 2022) shares ARC's input space but presents simpler problems with 20×20 images containing three distinct 3×3 objects. The ConceptARC dataset presents a set of manually crafted tasks, grouped and categorized by 16 distinct core concepts (Moskvichev et al., 2023). The PCFG benchmark (Mirchandani et al., 2023) is similar to our 1D-ARC and was generated using the probabilistic context-free grammar

LLM for the ARC In a recent study (Moskvichev et al., 2023), the capabilities of both automated methods and human cognition were explored with respect to the ARC. Their research employed state-of-the-art ARC solvers (top quarks, 2020; de Miquel Bleier, 2020) and GPT-4 to tackle tasks originating from the ConceptARC dataset, comparing these solutions with those produced by humans. The approach to prompt GPT-4 was comparable to our few-shot direct-grid encoding method outlined in Section 2.2. This study revealed that GPT-4 lags significantly behind both the leading ARC solver and human performance, a finding that aligns with our own. Other recent studies also echo this finding (Mirchandani et al., 2023; Camposampiero et al., 2023). However, it is critical to highlight that, based on our investigations, the proficiency of GPT-4 on the ConceptARC dataset could potentially be enhanced by adopting an object-based representation in the prompting process.

While our paper focused on the effects of representation, a recent study has shown improvements in the performance of LLMs on the ARC by incorporating hypothesis search Wang et al. (2023). Additionally, language instructions from LARC (Acquaviva et al., 2022) were utilized to assist LLMs in solving the ARC, as demonstrated in another recent study (Pu and Alford, 2023). Although the improvement was notable, it was not exceptional, suggesting the presence of additional challenges which we discuss in Section 6.

6 Conclusion and Future work

We have explored the capabilities and limitations of the GPT LLM in solving ARC tasks seen as representatives of a certain kind of human-like intelligence. Our exploration started with a straightforward, grid-based textual encoding approach, which revealed that GPT struggles due to the non-sequential representation of complex objects in text. We then introduced the 1D-ARC, a simplified, single-dimensional version of the ARC. By reducing the task complexity and dimensionality, we aimed to make ARC tasks more approachable for LLMs. Our evaluations on the 1D-ARC indicated improvements in performance but also highlighted that simplification alone could not bridge all the gaps in GPT's reasoning processes. In the third phase of our exploration, we adopted an object-based approach, integrating an external tool, the ARGA framework, to assist in object abstraction. This led to significant improvements in GPT's problem-solving abilities, reaffirming the importance of structured, object-based representations in complex reasoning tasks.

Our research uncovers potential avenues for future exploration. For instance, edge information was not fully utilized by the LLM, suggesting that GPT-4 may not be capable of dealing with graphs when represented in text form. As we delve deeper into the possibilities of structured representations, we might consider introducing a "language" of transformations for LLMs to use in solving ARC tasks.

Sub-abilities of LLMs Our research has predominantly focused on the LLM's ability to solve ARC tasks comprehensively. However, it's important to recognize that solving an ARC task can be a multi-step process, encompassing distinct sub-abilities. These include the LLM's capacity to abstract and understand objects from given inputs (Ability to Abstract), to formulate reasoning based on the understood input (Ability to Reason), and to generate accurate outputs using the reasoning and input (Ability to Execute). Each step presents unique challenges for LLMs. By this definition, our study can be viewed as assisting LLMs with the "Ability to Abstract". The recent study by Pu and Alford introduced in Section 5 can be seen as helping LLMs with the "Ability to Reason", since the reasoning is provided to LLMs directly. In both of these cases, LLMs still face difficulties in completely solving ARC tasks, which suggests that all three sub-abilities require further improvements.

Therefore, we believe that further experimentation aimed at dissecting and examining these sub-abilities could provide deeper insights into LLMs and pave the way for more sophisticated and capable language models in the future.

Visual Inputs for LLMs Recently, advancements have been made in developing LLMs capable of processing image inputs. GPT-4V is one such example (OpenAI, 2023b). At the time of our research, the GPT-4V API was not available. We thus conducted experiments by manually prompting ChatGPT-4 with image attachments. Evaluating on the set of 50 tasks described in Section 2.3, we found that GPT-4V, at its current stage, is only able to solve 2 out of the 50 tasks; in contrast, GPT-4 solves 23 tasks and even LLAMA2-13B solves 6 tasks. An example of solving an ARC task with GPT-4V is depicted in Figure 9. More details about this experiment are provided in Appendix G. These findings demonstrate that GPT-4V with a multimodal visual and language input almost fails completely on the selected ARC tasks and significantly underperforms pure language-based LLMs, even pure language LLMs that are *much smaller* than GPT-4V. Hence, it appears there is room for significant improvement for multimodal LLMs to correctly reason about combined visual and language inputs in the context of ARC tasks, which should serve as a future challenge for LLMs like GPT-4V.